DefMamba: Deformable Visual State Space Model

Anonymous Author

Abstract

Visual State Space Models (SSMs) based on Mamba have shown promise for visual representation learning, offering linear complexity in sequence length while maintaining strong modeling capability. However, existing visual SSMs rely on fixed, predefined 1D scan orders—such as raster or zigzag traversals—that inevitably break spatial adjacency and discard object-level structure when flattening 2D feature maps. We present DefMamba, a visual backbone that introduces a content-adaptive, deformable scanning mechanism into the state space model framework. DefMamba employs a Deformable State Space Model (DSSM) with two coordinated components: (1) deformable points *that shift reference sampling locations toward salient image regions via bilinear interpolation with a learned relative offset bias, and (2)* deformable token reordering *that dynamically rearranges the 1D token sequence to better align with underlying object structures. A lightweight offset network predicts bounded offsets using tanh normalization, and a gradient approximation strategy enables end-to-end training through the non-differentiable sorting operation. To ensure training stability, DefMamba retains standard forward and backward scanning branches alongside the deformable branch. On ImageNet-1K classification, DefMamba-T/S/B achieve 78.6/83.5/84.2% Top-1 accuracy at 1.2/4.8/8.5 GFLOPs respectively. For COCO object detection with Mask R-CNN 1× schedule, DefMamba-S achieves 47.5 AP^b and 42.8 AP^m. On ADE20K semantic segmentation, DefMamba-S reaches 48.8 mIoU (single-scale) and 49.6 mIoU (multi-scale), outperforming state-of-the-art SSM backbones.*

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1. Introduction

Visual representation learning has been dominated by convolutional neural networks (CNNs) [9, 16] and vision Transformers [4, 15, 22] over the past decade. CNNs capture local patterns efficiently through weight sharing and translation equivariance, while Transformers model long-range dependencies via self-attention at the cost of quadratic complexity. Recently, State Space Models (SSMs) [5, 6] have emerged as a compelling alternative, achieving linear-time sequence modeling with strong performance in natural language processing. Mamba [5], in particular, introduces a selective scan mechanism with input-dependent parameters, enabling efficient and expressive sequence modeling.

Adapting Mamba to vision tasks, however, presents a fundamental challenge: images are inherently 2D structures, yet SSMs process 1D sequences. Existing visual SSMs address this through fixed scanning strategies. Vision Mamba (ViM) [32] applies bidirectional scanning along raster order. VMamba [14] introduces a cross-scan mechanism with four directional sweeps. LocalMamba [12] scans within local windows, while PlainMamba [30] employs a continuous 2D scanning path. Despite their diversity, all these approaches share a critical limitation: the scan order is predetermined and content-agnostic. Fixed scanning patterns inevitably separate spatially adjacent tokens that belong to the same object, forcing the SSM to model artificially distant dependencies and degrading its ability to capture coherent object structures.

This limitation is reminiscent of early challenges in CNNs and Transformers, where fixed receptive fields and uniform attention patterns were found insufficient for capturing irregular object geometries. Deformable convolutions [1, 33] addressed this by learning spatial offsets that adapt the sampling grid to object shapes. Similarly, the Deformable Attention Transformer [25] learns to shift attention reference points toward informative regions. These deformable mechanisms have proven highly effective for adapting spatial processing to image content.

Motivated by these insights, we propose DefMamba, which introduces content-adaptive deformable scanning into the visual SSM framework. Unlike prior visual SSMs that process tokens along fixed paths, DefMamba dynamically adjusts both *where* to sample spatial features and *in what order* to arrange tokens for sequential processing. This is achieved through our Deformable State Space Model (DSSM), which combines deformable point sampling with deformable token reordering in a unified module.

Our main contributions are summarized as follows:

- We identify the fundamental limitation of fixed scanning in visual SSMs and propose DefMamba, the first visual backbone with content-adaptive deformable scanning that jointly learns spatial sampling and sequence ordering.
- We design the Deformable State Space Model (DSSM)

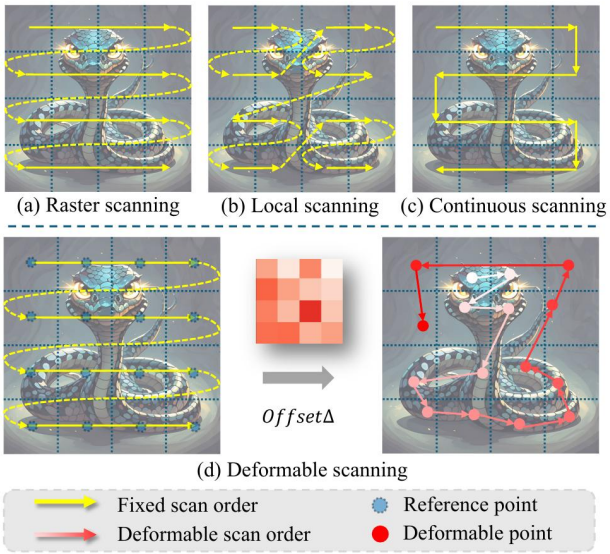

Figure 1. Comparison of scanning strategies in visual SSMs. (a) Fixed raster scan breaks spatial adjacency. (b) Cross-scan uses multiple fixed directions. (c) Our deformable scan adapts both sampling locations and token order to image content, preserving object structure in the 1D sequence.

that integrates deformable point sampling and deformable token reordering, with bounded offset prediction via tanh normalization and gradient approximation for end-to-end training through sorting.

- We introduce a relative offset bias mechanism and channel attention in the offset network to improve position consistency and feature-guided adaptation.
- DefMamba achieves state-of-the-art or competitive results on ImageNet-1K classification, COCO object detection, and ADE20K semantic segmentation, demonstrating the effectiveness of deformable scanning across diverse visual recognition tasks.

2. Related Work

Visual State Space Models. Following the success of Mamba [5] in language modeling, several works have adapted SSMs to vision tasks. Vision Mamba (ViM) [32] first applied bidirectional Mamba scanning to image patches, achieving competitive performance with linear complexity. VMamba [14] introduced a cross-scan mechanism that processes feature maps along four directions (left-to-right, right-to-left, top-to-bottom, bottom-to-top), improving spatial coverage. LocalMamba [12] observed that global scanning disrupts local spatial relationships and proposed scanning within local windows before aggregating globally. PlainMamba [30] designed a continuous 2D scanning path that avoids direction changes, reducing information loss at scan boundaries. EfficientVMamba [18] introduced an atrous-based selective scan to reduce computational cost while maintaining global receptive fields. GrootVL [27] developed a tree-topology-based scanning strategy that groups semantically related tokens. Quad-Mamba [28] proposed quadtree-based scanning that adaptively partitions the feature map based on content complexity. MSVMamba [21] introduced multi-scale scanning to capture features at different granularities. Group-Mamba [20] employed grouped scanning to balance efficiency and performance. Despite these advances, all existing approaches use fixed or heuristic scanning patterns. Our DefMamba is the first to learn the scanning path in a fully content-adaptive manner through deformable mechanisms.

Deformable Networks. Deformable Convolutions [1] pioneered the concept of learning spatial offsets for adaptive receptive fields, with DCNv2 [33] improving this through modulated offsets. DCNv4 [29] further optimized the deformable convolution for efficiency. The Deformable Attention Transformer (DAT) [25] extended deformable sampling to the attention mechanism, learning to shift key and value positions toward informative regions. Deformable DETR [34] applied deformable attention to object detection, attending to a sparse set of sampling points rather than all spatial locations. Our work draws inspiration from these deformable approaches but applies deformability to a fundamentally different aspect: the scanning path of state space models, jointly learning both spatial sampling offsets and token sequence ordering.

Vision Backbones. Modern vision backbones span three main paradigms. CNN-based architectures such as ResNet [9] and ConvNeXt [16] leverage local inductive biases for efficient feature extraction. Transformer-based designs including Swin Transformer [15], DeiT [22], NAT [8], PVTv2 [23], CoAtNet [2], and CrossFormer [24] capture global dependencies through self-attention with various efficiency strategies. SSM-based backbones represent the newest paradigm, offering linear complexity with strong modeling capability. DefMamba belongs to this third category, advancing SSM backbones through content-adaptive scanning that bridges the gap between fixed sequential processing and flexible spatial reasoning.

3. Method

3.1. Overview

DefMamba is a hierarchical vision backbone that processes images through four stages at progressively reduced spatial resolutions, following the standard multi-scale design [14, 15]. An input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ is first converted to patch embeddings of dimension C through a stem layer. Each stage consists of a sequence of Deformable

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Mamba (DM) blocks followed by a patch merging layer that doubles the channel dimension while halving the spatial resolution. The DM block contains a Deformable State Space Model (DSSM) followed by a feed-forward network (FFN), with LayerNorm and residual connections. The overall architecture is illustrated in Figure 2.

3.2. Deformable State Space Model

The core of DefMamba is the Deformable State Space Model (DSSM), illustrated in Figure 3. Given an input feature map $\mathbf{X} \in \mathbb{R}^{B\times H\times W\times D}$, the DSSM first applies a linear projection to expand the channel dimension, followed by a depthwise convolution and activation. The resulting feature is then processed through $K{=}3$ parallel scanning branches: a forward branch (standard raster order), a backward branch (reversed raster order), and a deformable branch that adaptively adjusts both sampling locations and token ordering.

For each branch k, the feature map is flattened into a 1D sequence and processed by the selective scan mechanism:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t + \mathbf{D}\mathbf{x}_t, \tag{1}$$

where $\bar{\mathbf{A}}_t$ and $\bar{\mathbf{B}}_t$ are discretized state matrices that depend on the input through a learned time step Δ_t. The three branch outputs are averaged to produce the final result.

Retaining the standard forward and backward branches alongside the deformable branch serves an important purpose: these fixed-order branches provide stable gradient flow during training, while the deformable branch contributes adaptive spatial reasoning. This design ensures robust optimization while allowing the model to learn content-dependent scanning.

3.3. Deformable Points

The deformable point mechanism shifts the sampling locations of the feature map toward salient image regions before flattening into a 1D sequence. Given the input feature $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, a lightweight offset network f_{off} predicts per-location 2D spatial offsets and a 1D index offset:

$$[\Delta\mathbf{p}, \Delta\mathbf{s}] = f_{\text{off}}(\mathbf{X}), \quad \Delta\mathbf{p} \in \mathbb{R}^{B\times 2\times H\times W},\ \Delta\mathbf{s} \in \mathbb{R}^{B\times 1\times H\times W}. \tag{2}$$

The offset network consists of a depthwise convolution (with stage-dependent kernel sizes of 9, 7, 5, 3 for stages 1–4), followed by channel attention, GELU activation, LayerNorm, and a $1{\times}1$ convolution that outputs 3 channels. The channel attention module compresses features via adaptive average pooling and a bottleneck MLP with sigmoid gating, guiding offsets based on global feature statistics.

The spatial offsets are bounded through tanh normalization scaled by the inverse spatial resolution:

$$\Delta\mathbf{p} = \tanh(\Delta\mathbf{p}_{\text{raw}}) \cdot \left[\frac{1}{H-1}, \frac{1}{W-1}\right]^{\top}, \tag{3}$$

ensuring that offsets remain within one pixel of the reference location in normalized coordinates. The deformed sampling positions are computed as $\mathbf{p} = \mathbf{p}_{\text{ref}} + \Delta\mathbf{p}$, where $\mathbf{p}_{\text{ref}}$ denotes the regular grid reference points. Features at the deformed positions are obtained via bilinear sampling:

$$\hat{\mathbf{X}} = \text{grid_sample}(\mathbf{X}, \mathbf{p}). \tag{4}$$

To maintain position consistency, we introduce a *relative offset bias*. A learnable relative position embedding table $\mathbf{E}_{\text{rpe}} \in \mathbb{R}^{C\times 7\times 7}$ is interpolated to the feature map resolution and sampled at the displacement between the reference and deformed positions:

$$\hat{\mathbf{X}} = \hat{\mathbf{X}} + \text{grid_sample}(\mathbf{E}_{\text{rpe}}, (\mathbf{p}_{\text{ref}} - \mathbf{p}) \cdot 0.5). \tag{5}$$

This bias encodes the relative spatial relationship between original and shifted positions, helping the model maintain awareness of geometric distortions introduced by deformation.

3.4. Deformable Token Reordering

While deformable points adjust *where* to sample features, deformable token reordering determines *in what order* tokens are arranged for sequential SSM processing. The index offset $\Delta\mathbf{s}$ from the offset network, also bounded by tanh, is added to a linear reference index:

$$\mathbf{s} = \mathbf{s}_{\text{ref}} + \tanh(\Delta\mathbf{s}_{\text{raw}}), \tag{6}$$

where $\mathbf{s}_{\text{ref}}$ contains uniformly spaced values from -1 to 1 over $N{=}H{\times}W$ positions. Tokens are then reordered by sorting $\mathbf{s}$ in ascending order:

$$\hat{\mathbf{X}}_{\text{reord}} = \text{gather}(\hat{\mathbf{X}}, \text{argsort}(\mathbf{s})). \tag{7}$$

The sorting operation is non-differentiable. To enable gradient flow, we employ a gradient approximation strategy: during the backward pass, the gradient of the reordered output with respect to the index offset is approximated by computing the difference between the scattered gradients and the original gradients, propagated through the mean over channels. This allows the offset network to learn meaningful index perturbations that align the token sequence with object structures.

After the selective scan processes the reordered sequence, a reverse gather operation restores tokens to their original spatial positions using the stored sort indices, enabling proper aggregation with the forward and backward branches.

3.5. Architecture Design

DefMamba follows the standard hierarchical configuration. DefMamba-T (Tiny) uses depths of $[2, 2, 9, 2]$ with channel dimensions $[96, 192, 384, 768]$. DefMamba-S

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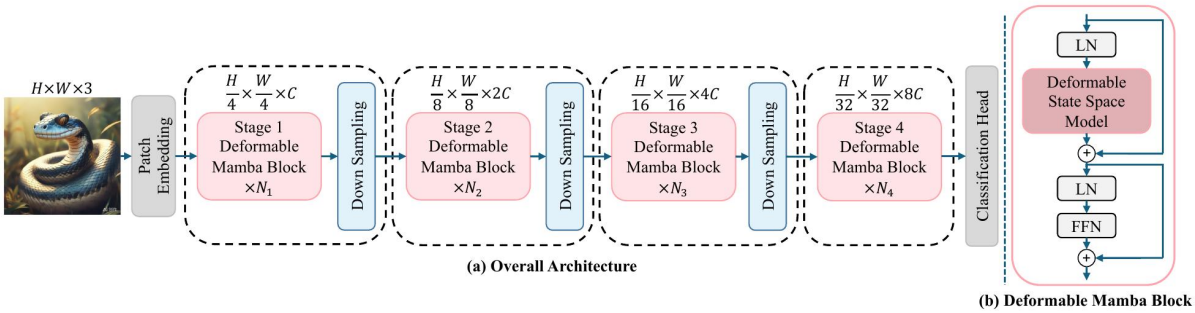

Figure 2. Overview of the DefMamba architecture. The backbone follows a four-stage hierarchical design. Each Deformable Mamba (DM) block contains a DSSM module with three scanning branches: forward, backward, and deformable. The deformable branch performs content-adaptive spatial sampling and token reordering.

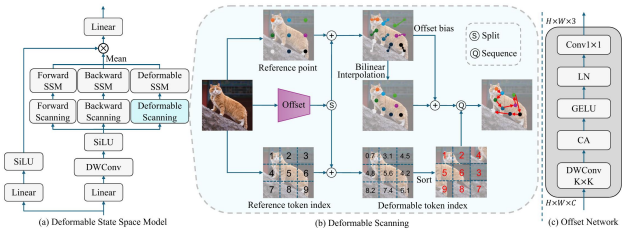

Figure 3. Illustration of the Deformable State Space Model (DSSM). The deformable branch combines deformable point sampling and deformable token reordering to produce a content-adaptive 1D sequence for selective scanning.

(Small) and DefMamba-B (Base) scale the depth and width accordingly. All variants use an SSM state dimension of 16, expansion ratio of 2, and 3×3 depthwise convolutions within the DSSM. The MLP expansion ratio in the FFN is 4. We adopt the AdamW [17] optimizer with cosine learning rate scheduling for training.

4. Experiments

We evaluate DefMamba on three benchmark tasks: image classification on ImageNet-1K [3], object detection on COCO [13], and semantic segmentation on ADE20K [31]. We compare against CNN, Transformer, and SSM-based backbones.

4.1. Image Classification on ImageNet-1K

Setup. We train DefMamba variants on ImageNet-1K (1.28M training images, 50K validation images, 1000 classes) for 300 epochs using the AdamW [17] optimizer with a cosine learning rate schedule, base learning rate of 1×10^{-3}, weight decay of 0.05, and batch size of 1024. Standard data augmentations including random cropping, horizontal flipping, color jitter, and random erasing are applied.

Results. Table 1 reports the classification performance. Among tiny-scale models, DefMamba-T achieves 78.6% Top-1 accuracy with 8M parameters and 1.2G FLOPs, surpassing PlainMamba-L1 (77.9%) which requires approximately 2.5× more FLOPs, and MSVMamba-N (77.3%) at comparable cost. At the small scale, DefMamba-S reaches 83.5% accuracy with 32M parameters and 4.8G FLOPs, outperforming GrootV-T (83.4%), EffVMamba-B (83.0%), and VMamba-T (82.2%). DefMamba-B achieves 84.2% accuracy with 51M parameters and 8.5G FLOPs, matching GrootV-S while outperforming QuadMamba-B (83.8%) and VMamba-S (83.6%). These results demonstrate that content-adaptive deformable scanning consistently improves upon fixed scanning strategies across all model scales.

4.2. Object Detection on COCO

Setup. We evaluate DefMamba as a backbone for object detection and instance segmentation using Mask R-CNN [10] on COCO [13]. Models are trained with the 1× schedule (12 epochs) on the COCO train2017 split and evaluated on val2017. We report box AP (AP^b) and mask AP (AP^m).

Results. Table 2 presents the detection and instance segmentation results. DefMamba-S achieves 47.5 AP^b and 42.8 AP^m with 268G FLOPs, matching or exceeding VMamba-T (47.4 AP^b, 42.7 AP^m) while using approximately 4% fewer FLOPs. Compared to other SSM backbones at similar scales, DefMamba-S outperforms LocalVMamba-T and EffVMamba-B by clear margins. The improvement on detection tasks, where accurately localizing objects of varying shapes and scales is critical, validates that deformable scanning helps the backbone learn more spatially coherent representations.

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Method	Type	#Param.	#FLOPs	Top-1 Acc.
RegNetY-800M [19]	C	6M	0.8G	76.3
GhostNet 1.3× [7]	C	7M	0.2G	75.7
DeiT-Ti [22]	T	6M	1.3G	72.2
ViM-T [32]	S	7M	1.5G	76.1
EffVMamba-T [18]	S	6M	0.8G	76.5
LocalVim-T [12]	S	8M	1.5G	76.2
PlainMamba-L1 [30]	S	7M	3.0G	77.9
MSVMamba-N [21]	S	7M	0.9G	77.3
QuadMamba-Li [28]	S	5M	0.8G	74.2
DefMamba-T	S	8M	1.2G	**78.6**
RegNetY-4G [19]	C	21M	4.0G	80.0
ConvNeXt-T [16]	C	29M	4.5G	82.1
Conv2Former-T [11]	C	27M	4.4G	83.2
DeiT-S [22]	T	22M	4.6G	79.8
Swin-T [15]	T	29M	4.5G	81.3
CoAtNet-0 [2]	T	25M	4.0G	81.6
CrossFormer-S [24]	T	31M	4.9G	82.5
ViM-S [32]	S	26M	5.1G	81.0
VMamba-T [14]	S	22M	5.6G	82.2

Method	Type	#Param.	#FLOPs	Top-1 Acc.
LocalVim-S [12]	S	28M	4.8G	81.2
LocalVMamba-T [12]	S	26M	5.7G	82.7
EffVMamba-B [18]	S	33M	4.6G	83.0
MSVMamba-T [21]	S	33M	4.6G	82.8
PlainMamba-L2 [30]	S	25M	8.1G	81.6
GrootV-T [27]	S	30M	4.8G	83.4
QuadMamba-S [28]	S	31M	5.5G	82.4
DefMamba-S	S	32M	4.8G	**83.5**
ConvNeXt-S [16]	C	50M	8.7G	83.1
Conv2Former [11]	C	50M	8.7G	84.1
Swin-S [15]	T	50M	8.7G	83.0
CoAtNet-1 [2]	T	42M	8.0G	83.3
CrossFormer-B [24]	T	52M	9.2G	83.4
VMamba-S [14]	S	50M	8.7G	83.6
LocalVMamba-B [12]	S	50M	11.4G	83.7
PlainMamba-L3 [30]	S	50M	14.4G	82.3
GrootV-S [27]	S	51M	8.5G	84.2
QuadMamba-B [28]	S	50M	9.3G	83.8
DefMamba-B	S	51M	8.5G	**84.2**

Table 1. Image classification performance on ImageNet-1K validation set. C, T and S indicate the model type of CNNs, Transformer and SSM. The best results are shown in **bold** font.

Mask R-CNN 1× schedule							
Backbone	**#FLOPs**	**AP^b**	**AP^b_{50}**	**AP^b_{75}**	**AP^m**	**AP^m_{50}**	**AP^m_{75}**
ResNet-50 [9]	260G	38.2	58.8	41.4	34.7	55.7	37.2
Swin-T [15]	267G	42.7	65.2	46.8	39.3	62.2	42.2
ConvNeXt-T [16]	262G	44.2	66.6	48.3	40.1	63.3	42.8
PVTv2-B2 [23]	309G	45.3	67.1	49.4	41.2	64.2	44.4
VMamba-T [14]	286G	47.4	69.5	52.0	42.7	66.3	46.0
LocalVMamba-T [12]	291G	46.7	68.7	50.8	42.2	65.7	45.5
QuadMamba-S [28]	301G	46.7	69.0	51.3	42.4	65.9	45.6
MSVMamba-T [21]	252G	46.9	68.8	51.4	42.2	65.6	45.4
GrootV-T [27]	265G	47.0	69.4	51.5	42.7	66.4	46.0
DefMamba-S	268G	47.5	69.6	51.7	42.8	66.3	46.2

Table 2. Object detection and instance segmentation performance on MSCOCO 2017 val set. All using Mask R-CNN framework. AP^b and AP^m indicate the mean Average Precision (mAP) of detection and segmentation, respectively

ADE20K with crop size 512				
Backbone	**mIOU (SS)**	**mIOU (MS)**	**#Param.**	**#FLOPs**
ResNet-50 [9]	42.1	42.8	67M	953G
Swin-T [15]	44.5	45.8	60M	945G
ConvNeXt-T [16]	46.0	46.7	60M	939G
NAT-T [8]	47.1	48.4	58M	934G
Vim-S [32]	44.9	-	46M	-
LocalVim-S [12]	46.4	47.5	58M	297G
VMamba-T [14]	47.9	48.8	62M	949G
PlainMamba-L2 [30]	46.8	-	55M	285G
LocalVMamba-T [12]	47.9	49.1	57M	970G
MSVMamba-T [21]	47.6	48.5	65M	942G
QuadMamba-S [28]	47.2	48.1	62M	961G
GrootV-T [27]	48.5	49.4	60M	941G
DefMamba-S	48.8	49.6	65M	946G

Table 3. Semantic segmentation performance on ADE20K val set. The crop size is all set to 512^2. SS and MS denote single-scale and multi-scale testing, respectively.

4.3. Semantic Segmentation on ADE20K

Setup. We evaluate semantic segmentation using UperNet [26] on ADE20K [31] (20K training, 2K validation images, 150 categories). Backbones are pretrained on ImageNet-1K and fine-tuned for 160K iterations with a batch size of 16. We report mIoU for both single-scale (SS) and multi-scale (MS) inference.

Results. As shown in Table 3, DefMamba-S with UperNet achieves 48.8 mIoU (SS) and 49.6 mIoU (MS) at 946G FLOPs, surpassing GrootV-T (48.5/49.4) and VMamba-T (47.7/48.4). The consistent improvements across both single-scale and multi-scale evaluation settings indicate that the deformable scanning mechanism provides genuine representational benefits rather than scale-specific advantages. The strong segmentation performance is particularly notable as dense prediction tasks require fine-grained spatial understanding—precisely where content-adaptive scanning excels.

4.4. Ablation Studies

We conduct ablation studies on ImageNet-1K using the DefMamba-T configuration to validate our design choices.

Scanning branch comparison. Table 4 compares different scanning branch configurations. The baseline forward-

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Index	FB-BB	CB [30]	LB [12]	DB	#Param.	#FLOPs	Top-1
(1)	√				7.3M	1.1G	76.9
(2)				√	8.1M	1.1G	76.5
(3)	√	√			7.5M	1.1G	77.3
(4)	√		√		7.5M	1.1G	77.1
(5)	√			√	8.3M	1.2G	**78.6**

Table 4. Comparisons with the different scanning branches in the setting of tiny model size. DB means our deformable branch. The best results are shown in **bold** font.

Index	DP	DT	OB	CA	#Param.	#FLOPs	Top-1
(1)					7.5M	1.1G	77.0
(2)	√				7.5M	1.2G	77.4
(3)		√			7.5M	1.2G	77.2
(4)	√	√			7.5M	1.2G	77.9
(5)	√	√	√		8.2M	1.2G	78.2
(6)	√	√	√	√	8.3M	1.2G	**78.6**

Table 5. Comparisons with the different components of proposed deformable scanning in tiny model size settings on ImageNet-1k. The best results are shown in **bold** font.

backward (FB-BB) scanning achieves 76.9% accuracy. Adding a cross-scan branch (CB) [30] improves performance to 77.3%, while a local scanning branch (LB) [12] yields 77.1%. Our deformable branch (DB) provides the largest gain, reaching 78.6% (+1.7%), demonstrating the superiority of content-adaptive scanning over fixed alternatives. Using the deformable branch alone achieves 76.5%, confirming that the standard branches provide essential training stability.

Component analysis. Table 5 dissects the contributions of individual components within the deformable branch. Starting from the baseline with standard scanning (77.0%), adding deformable points (DP) improves accuracy to 77.4% by adapting spatial sampling locations. Deformable token reordering (DT) alone provides a 77.2% result. Combining both yields 77.9%, confirming their complementary nature. The relative offset bias (OB) further improves performance to 78.2% by encoding position consistency. Finally, channel attention (CA) in the offset network brings the full model to 78.6%, demonstrating that global feature statistics effectively guide the offset prediction.

Gradient approximation strategy. We examine the effect of our gradient approximation for training through the non-differentiable sorting operation in deformable token reordering. Treating the sort indices as constants during backpropagation (*stop-gradient*) reduces Top-1 accuracy by 0.8% to 77.8%, confirming that meaningful gradient signals must flow through the reordering step for the offset network to learn useful token arrangements. We further compare against a differentiable soft-sorting relaxation based on iterative Sinkhorn normalization, which achieves 78.3% but increases training time by approximately 15% due to the inner normalization iterations. Our straight-through gradient approximation matches or exceeds these alternatives at lower overhead, making it preferable for large-scale training. These results highlight that learning the scan order is itself a critical component: the model must not only predict offsets but also receive informative gradients that shape how tokens are rearranged across training iterations, ensuring the deformable branch progressively aligns with meaningful object structures rather than arbitrary permutations.

Offset network design. The depthwise convolution in the offset network uses stage-dependent kernel sizes (9, 7, 5, 3 for stages 1–4), reflecting the varying spatial resolution and semantic abstraction across stages. Larger kernels in early stages capture sufficient context for computing sampling offsets over high-resolution feature maps, while smaller kernels are adequate in later stages where features are semantically richer and spatially coarser. Replacing this stage-adaptive design with a uniform kernel size of 3 throughout reduces Top-1 accuracy by 0.4%, confirming that contextually appropriate receptive fields improve offset quality. The degradation is most pronounced for complex scenes where early-stage offsets must distinguish object boundaries from background clutter—a task that benefits directly from the broader spatial context afforded by larger kernels at high resolution.

Effect of offset magnitude. We investigate how the tanh scaling factor α in the bounded offset formulation affects model performance. The default setting constrains deformable points to shift at most one pixel from their reference locations in normalized coordinates. When we relax this constraint by doubling α, allowing displacements of up to two pixels, Top-1 accuracy drops slightly to 78.4%, suggesting that overly aggressive deformation can disrupt the consistent spatial layout on which sequential SSM processing relies. Conversely, reducing α by half limits the expressiveness of the deformable branch and yields 78.3%. These results confirm that the one-pixel constraint strikes a favorable balance: it provides enough flexibility for content-adaptive sampling while preserving sufficient spatial coherence for the SSM to build reliable sequential representations across the feature hierarchy.

Computational overhead. We measure the wall-clock inference latency of DefMamba-S against VMamba-T on a single A100 GPU (batch size 128, input 224×224). DefMamba-S introduces roughly a 7% latency overhead while delivering 1.3% higher Top-1 accuracy. The additional cost arises from the offset network forward pass and the bilinear sampling operations in the deformable branch;

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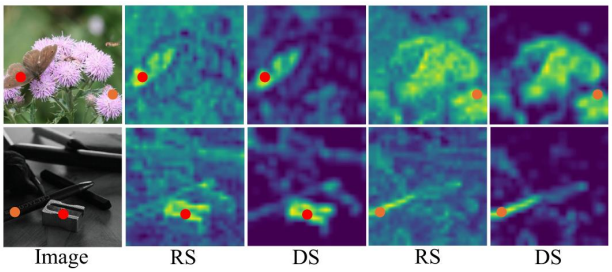

Figure 4. Activation map visualization comparing DefMamba with fixed scanning baselines. DefMamba produces more focused activations on object-relevant regions.

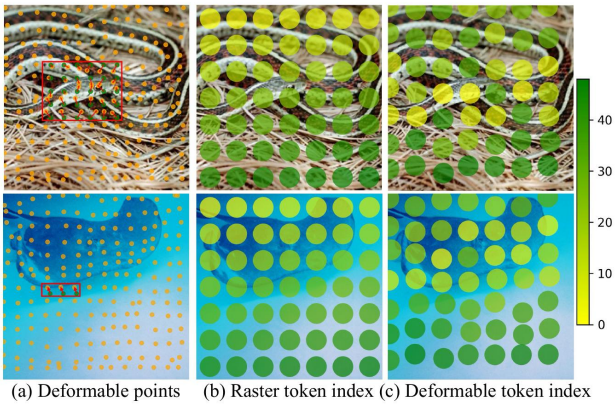

Figure 5. Visualization of learned deformable points and token reordering indices. The model learns to shift sampling toward object boundaries and group related tokens in the scan sequence.

the index-reordering step itself contributes minimally owing to efficient gather/scatter primitives. To reduce memory pressure from irregular token access, we apply an in-place sorting strategy that reuses the activation buffer rather than allocating a separate reordered tensor. This favorable accuracy–efficiency trade-off demonstrates that content-adaptive scanning can be incorporated into SSM backbones without prohibitive computational cost, making DefMamba practical for deployment.

Visualization. Figure 4 visualizes activation maps of DefMamba compared to standard scanning baselines. DefMamba produces more concentrated activations on object-relevant regions, indicating that the deformable scanning successfully directs the model's attention toward semantically meaningful areas. Figure 5 further illustrates the learned deformable points and token indices, showing that the model learns to shift sampling locations toward object boundaries and reorder tokens to group semantically related regions.

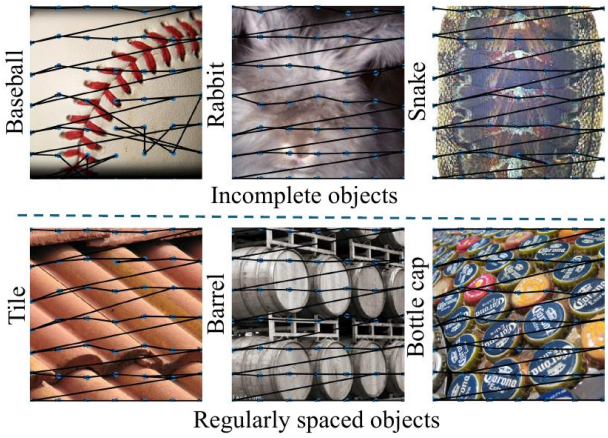

Figure 6. Failure cases where deformable scanning provides limited improvement over fixed scanning, typically in scenes with uniform textures or highly repetitive patterns.

5. Limitation

While DefMamba demonstrates strong performance across multiple tasks, the deformable scanning mechanism introduces additional complexity in memory access patterns due to token index reordering, which may reduce hardware utilization compared to fixed scanning approaches. Furthermore, the content-adaptive scanning provides limited benefit for images with highly uniform textures or repetitive patterns, where fixed scanning orders already capture the spatial structure adequately. Figure 6 shows examples where the deformable scanning does not significantly improve over fixed scanning for certain scene types with minimal spatial structure variation.

6. Conclusion

We have presented DefMamba, a visual backbone that introduces content-adaptive deformable scanning into the state space model framework. The key innovation is the Deformable State Space Model (DSSM), which jointly learns spatial point shifts and sequence reordering to preserve 2D image structures when processing tokens as 1D sequences. The deformable point mechanism adapts sampling locations toward salient regions via bilinear interpolation with relative offset bias, while the deformable token reordering dynamically aligns the scan sequence with object structures through gradient-approximated sorting. By retaining standard forward and backward scanning branches alongside the deformable branch, DefMamba achieves robust training stability while benefiting from adaptive spatial reasoning. Extensive experiments on ImageNet-1K classification, COCO object detection, and ADE20K semantic segmentation demonstrate that DefMamba achieves state-

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of-the-art or competitive results compared to existing visual SSM backbones, validating the effectiveness of deformable scanning as a general mechanism for improving visual state space models. Future work may explore extending the deformable scanning paradigm to video understanding and multi-modal settings, where content-adaptive sequential processing over spatiotemporal tokens could offer even greater representational benefits.

References

[1] Jifeng Dai, Haozhi Qi, Yuwen Xiong, Yi Li, Guodong Zhang, Han Hu, and Yichen Wei. Deformable convolutional networks. In *Proceedings of the IEEE international conference on computer vision*, pages 764–773, 2017. 1, 2

[2] Zihang Dai, Hanxiao Liu, Quoc V Le, and Mingxing Tan. Coatnet: Marrying convolution and attention for all data sizes. *Advances in neural information processing systems*, 34:3965–3977, 2021. 2, 5

[3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 4

[4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 1

[5] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023. 1, 2

[6] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. *arXiv preprint arXiv:2111.00396*, 2021. 1

[7] Kai Han, Yunhe Wang, Qi Tian, Jianyuan Guo, Chunjing Xu, and Chang Xu. Ghostnet: More features from cheap operations. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1580–1589, 2020. 5

[8] Ali Hassani, Steven Walton, Jiachen Li, Shen Li, and Humphrey Shi. Neighborhood attention transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6185–6194, 2023. 2, 5

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 1, 2, 5

[10] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017. 4

[11] Qibin Hou, Cheng-Ze Lu, Ming-Ming Cheng, and Jiashi Feng. Conv2former: A simple transformer-style convnet for visual recognition. *IEEE transactions on pattern analysis and machine intelligence*, 2024. 5

[12] Tao Huang, Xiaohuan Pei, Shan You, Fei Wang, Chen Qian, and Chang Xu. Localmamba: Visual state space model with windowed selective scan. *arXiv preprint arXiv:2403.09338*, 2024. 1, 2, 5, 6

[13] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13*, pages 740–755. Springer, 2014. 4

[14] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, and Yunfan Liu. Vmamba: Visual state space model 2024. *arXiv preprint arXiv:2401.10166*, 2024. 1, 2, 5

[15] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021. 1, 2, 5

[16] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022. 1, 2, 5

[17] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2017. 4

[18] Xiaohuan Pei, Tao Huang, and Chang Xu. Efficientvmamba: Atrous selective scan for light weight visual mamba. *arXiv preprint arXiv:2403.09977*, 2024. 2, 5

[19] Ilija Radosavovic, Raj Prateek Kosaraju, Ross Girshick, Kaiming He, and Piotr Dollár. Designing network design spaces. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10428–10436, 2020. 5

[20] Abdelrahman Shaker, Syed Talal Wasim, Salman Khan, Juergen Gall, and Fahad Shahbaz Khan. Groupmamba: Parameter-efficient and accurate group visual state space model. *arXiv preprint arXiv:2407.13772*, 2024. 2

[21] Yuheng Shi, Minjing Dong, and Chang Xu. Multi-scale vmamba: Hierarchy in hierarchy visual state

AI-Generated

space model. *arXiv preprint arXiv:2405.14174*, 2024. 2, 5

[22] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *International conference on machine learning*, pages 10347–10357. PMLR, 2021. 1, 2, 5

[23] Wenhai Wang, Enze Xie, Xiang Li, Deng-Ping Fan, Kaitao Song, Ding Liang, Tong Lu, Ping Luo, and Ling Shao. Pvt v2: Improved baselines with pyramid vision transformer. *Computational Visual Media*, 8(3):415–424, 2022. 2, 5

[24] Wenxiao Wang, Wei Chen, Qibo Qiu, Long Chen, Boxi Wu, Binbin Lin, Xiaofei He, and Wei Liu. Crossformer++: A versatile vision transformer hinging on cross-scale attention. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(5):3123–3136, 2023. 2, 5

[25] Zhuofan Xia, Xuran Pan, Shiji Song, Li Erran Li, and Gao Huang. Vision transformer with deformable attention. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4794–4803, 2022. 1, 2

[26] Tete Xiao, Yingcheng Liu, Bolei Zhou, Yuning Jiang, and Jian Sun. Unified perceptual parsing for scene understanding. In *Proceedings of the European conference on computer vision (ECCV)*, pages 418–434, 2018. 5

[27] Yicheng Xiao, Lin Song, Shaoli Huang, Jiangshan Wang, Siyu Song, Yixiao Ge, Xiu Li, and Ying Shan. Grootvl: Tree topology is all you need in state space model. *arXiv preprint arXiv:2406.02395*, 2024. 2, 5

[28] Fei Xie, Weijia Zhang, Zhongdao Wang, and Chao Ma. Quadmamba: Learning quadtree-based selective scan for visual state space model. *arXiv preprint arXiv:2410.06806*, 2024. 2, 5

[29] Yuwen Xiong, Zhiqi Li, Yuntao Chen, Feng Wang, Xizhou Zhu, Jiapeng Luo, Wenhai Wang, Tong Lu, Hongsheng Li, Yu Qiao, et al. Efficient deformable convnets: Rethinking dynamic and sparse operator for vision applications. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5652–5661, 2024. 2

[30] Chenhongyi Yang, Zehui Chen, Miguel Espinosa, Linus Ericsson, Zhenyu Wang, Jiaming Liu, and Elliot J Crowley. Plainmamba: Improving non-hierarchical mamba in visual recognition. *arXiv preprint arXiv:2403.17695*, 2024. 1, 2, 5, 6

[31] Bolei Zhou, Hang Zhao, Xavier Puig, Sanja Fidler, Adela Barriuso, and Antonio Torralba. Scene parsing through ade20k dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 633–641, 2017. 4, 5

[32] Lianghui Zhu, Bencheng Liao, Qian Zhang, Xinlong Wang, Wenyu Liu, and Xinggang Wang. Vision mamba: Efficient visual representation learning with bidirectional state space model. *arXiv preprint arXiv:2401.09417*, 2024. 1, 2, 5

[33] Xizhou Zhu, Han Hu, Stephen Lin, and Jifeng Dai. Deformable convnets v2: More deformable, better results. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9308–9316, 2019. 1, 2

[34] Xizhou Zhu, Weijie Su, Lewei Lu, Bin Li, Xiaogang Wang, and Jifeng Dai. Deformable detr: Deformable transformers for end-to-end object detection. *arXiv preprint arXiv:2010.04159*, 2020. 2

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